

Practical 6

15th September 2026

1. Using AL2CO server, obtain the positional conservation scores from multiple sequence alignment (MSA) of given set of protein sequences (set1 and set2) using the methods given below:

- (i) Unweighted frequency and entropy-based measure
- (ii) Unweighted frequency and variance-based measure
- (iii) Unweighted frequency and sum of pairs measure (HINT: Download and use BLOSUM62 scoring matrix)
- (iv) Weighted frequency and variance-based measure
- (v) Normalize the scores obtained with (i)

Sequences:

Set 1

P69905, P01946, P01942, P01966, P01958, P01959, P01965, P06635, P60529, P80043 and P01980

Set 2

TPIS_HUMAN, TPIS_YEAST, TPIS_GRAGA, TPIS_TRYCR, TPIS_MAIZE, TPIS_MOUSE, TPIS_DROME, TPIS_RABIT, TPIS_CAEEL

Instructions:

A. Access the lab Linux system through SSH (ssh username@ip)

(Ex. ssh bioinfo@10.21.50.138)

Username: bioinfo

Password: protein@321

IP address: 10.21.50.138

Command: al2co

Once you type the command (**al2co**) in a terminal, all options will be listed and can be used appropriately.

- B. Enter into the Dir created earlier (cd {roll_no}) else create Dir with your ROLL NUMBER (mkdir {roll_no})
- C. Edit the alignment file and remove the first line (CLUSTAL O(1.2.4) multiple sequence alignment).
- D. Copy the alignment file from local to remote using **scp** command: scp <source of file> <destination>
- E. Similarly, once done, copy back the results to your local system using **scp** command

Example: **scp** <location to your alignment file (eg: alignment.aln)> bt3040@10.21.49.164:~/<your folder name>/

Documentation for AL2CO is available at:

<https://github.com/TheApacheCats/al2co/blob/master/README>

2. For the same sequences as question 1 use AL2CO server, obtain the positional conservation scores from multiple sequence alignment (MSA) of given set of protein sequences (set1 and set2) using the methods given below:
 - (i) Unweighted frequency and entropy-based measure
 - (ii) Unweighted frequency and variance-based measure
 - (iii) Unweighted frequency and sum of pairs measure (HINT: Download and use BLOSUM62 scoring matrix)
 - (iv) Weighted frequency and variance-based measure
 - (v) Normalize the scores obtained with (i)

(<http://prodata.swmed.edu/al2co/al2co.php>)
3. Tabulate the topmost 10 residues with the highest and lowest conservation scores (in both Set1 and Set 2) obtained with method (i).
4. Write a program to compute the conservation score from MSA using unweighted frequency, and entropy, variance, and sum of pairs-based measures.
5. Using the program written in Q3 (unweighted frequency and entropy-based measure), compare the MSA from Clustal Omega, MAFFT, and MUSCLE. Identify the residues with (i) similar and (ii) Different conservation scores among the three alignment methods.
6. Check the scores manually at positions 9, 11, 20, 22, and 30 (use MSA from Clustal Omega)
7. Obtain the conservation score of 1BTM, A-chain using Consurf server (<https://consurf.tau.ac.il/>)

Deadline: 21st September 2026